



# Md. Babul Miah

## CLIMATE AND OCEAN PHYSICIST

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## RESEARCH INTEREST

- Seasonal to Decadal Climate Prediction
- Changes in Hydrological Cycle
- Air-Sea Interaction
- Atmospheric Circulation
- Land-Atmosphere Interaction
- Global Monsoon, Indian Ocean and Asian Monsoon
- Physics of Global Warming, Biogeochemical Modeling
- Extreme Events, Irreversibility of Climate Extreme
- Machine Learning for Earth System Science
- Climate Model Development, Numerical Climate Modeling

## SKILLS

- Tools and Languages:** Python, NCL, CDO, C, bash scripting, GrADS, Super Computing.
- Quantitative Research:** Data Analysis, Statistical Modeling, Timeseries analysis, Climate Strategies, CO2 increase impact mechanism & analysis, Climate Risk, Mathematical Modeling, Numerical Climate Modeling.
- Machine Learning:** Scikit-learn, PyTorch, Keras, TensorFlow.
- Climate Models:** WRF, GFDL-ESM4,

## EDUCATION

### Ph.D in Climate, Environment and Energy | Sept. 2023-present (Expected Feb. 2026)

*Jeonbuk National University, Republic of Korea*

- Relevant Coursework: Advanced Remote Sensing, Analysis of Earth System Data, Advance Geophysical Imaging, Advance Climate Technology, Advance Atmospheric Physic and Chemistry, Advance Hydrology, Advance Climate and Ocean Modeling
- CGPA: 4.00 (Out of 4.00)

### B.Sc (Hons) in Environmental Science and Disaster Management | 2017-2022

*Gopalganj Science and Technology University, Bangladesh*

- Relevant Coursework: Atmospheric Physics, Meteorology and Climate Process, Environmental Modelling, Remote Sensing, Global Warming and Climate Change, Geographic Information System(GIS). Environmental Mathematics.
- CGPA: 2.80 (Out of 4.00)

## PROFESSIONAL & RESEARCH EXPERIENCE

### Research Assistant | 2023-present

*Climate Modeling Lab, Jeonbuk National University, Republic of Korea*

- Investigate the Irreversibility of extreme precipitation of multiple carbon neutrality scenarios.
- Sensitivity of CO<sub>2</sub> increase in the atmosphere.
- Analyze observed satellite, multiple climate models to investigate climate change mechanism & influence.
- Run WRF model to investigate downscaling for NIMS\_UKESM1 multiple carbon neutrality output of precipitation.

### Assistant Director and Research Associate | 2020-present

*Research and Skill Development Society, Bangladesh ( <https://rsds-bd.org/assistant-director/> )*

*Climate and Weather Prediction Center, Bangladesh ( <https://rsds-bd.org/cwcr/> )*

- Collaborated with a research team to prediction of extreme rainfall indices using ocean atmospheric teleconnection.
- Conducted experiments using machine learning and deep learning techniques for extreme rainfall indices.
- Analyzed and interpreted climate data, contributing to a research paper submitted for publication

### Undergraduate Research Assistant | 2019-2020

*Dept. of Environmental Science and Disaster Management, Gopalganj Science and Technology University, Bangladesh*

- Use observed & satellite data to investigate who to predict extreme precipitation indices.
- Setup and & run Weather Research & Forecasting Model (WRF) to investigate tropical cyclones

## RELEVANT DEVELOPMENTS & ACTIVITIES

- Research Publications** – 2+ Peer-reviewed publications, & 7+ conference proceedings. [Google Scholar Link](#)
- Scientific Data Analysis and Visualization with Python**– A book for beginners in Python to plot climate data & do basic statistical analysis. Zenodo: <https://doi.org/10.5281/zenodo.7592894>
- Scientific Data Analysis and Visualization with SPSS** –A book for ideas a coherent and systematic exploration of the fundamental principles of computer data, statistical theory, and practical SPSS methods,. Zenodo: <https://doi.org/10.5281/zenodo.14195590>
- Python Data Analysis Tutorial** – from basics of using Python to Climate data analysis, applying statistical tool & plot, GIS & Remote Sensing, Climate Modeling tutorial, to finally applying machine learning. [page: https://coml-babul.org/video-lectures/](https://coml-babul.org/video-lectures/)

## PUBLICATIONS [COMPLETE LIST LINK]

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[4]. Miah, M.B., Park, J.Y., Lee, M.U., Jeon, W., Byun., Y.H., Sung., H.M., Hong.(2025). Irreversibility of East Asian Extreme Precipitation to Carbon-Neutrality Pathways: A WRF Dynamical Downscaling Study. **In Prepare**

[3]. Miah, M.B., Park, J.Y., Lee, M.U., Jeon, W., Byun., Y.H., Sung., H.M., Hong.(2025). Joint Return Period of Extreme Precipitation Under Multiple Carbon Neutrality Scenarios. **In Prepare**

[2]. **Miah, M.B.**, Park, J.Y., Lee, M.U., Jeon, W., Byun., Y.H., Sung., H.M., Uddin, M.J., Mondal S.K.(2025). Irreversibility of Extreme Precipitation Intensity in Global Monsoon Areas Under Multiple Carbon Neutrality Scenarios. <https://doi.org/10.1016/j.wace.2025.100843>

[1]. Uddin, M. J., Li, Y., Tamim, M. Y., **Miah, M. B.**, & Ahmed, S. M. S., (2022). Extreme rainfall indices prediction with atmospheric parameters and ocean-atmospheric teleconnections using Random Forest model. Journal of Applied Meteorology and Climatology. <https://doi.org/10.1175/jamc-d-21-0170.1>

## CONFERENCE PROCEEDINGS [SELECTED]

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[3]. **Miah, M.B.**, Park, J.Y., Lee, M.U., Jeon, W., Byun., Y.H., Sung., H.M., Hong, J.G, Uddin, M.J., Mondal S.K. (2025). Irreversibility of Extreme Precipitation in Global Monsoon Areas Under Multiple Carbon Neutrality Scenarios. American Geophysical Union(AGU) Annual Meeting (2025), New Orleans, LA 70130, USA

[2]. Eshita, N. R., Uddin, M. J., **Miah, Md. B.**, Ahmed, S. M. S., Rahman, Md. H. (2025). Recent summer heatwave in Bangladesh enhanced by Atlantic multi-decadal oscillation and large-scale atmospheric factors. 10<sup>th</sup> International Conference on Water and Flood Management-ICWFM, BUET, Dhaka, Bangladesh. <https://doi.org/10.5281/zenodo.15253705>

[1]. **Miah, M.B.**, Park, J.Y., Lee, M.U., Jeon, W., Byun., Y.H., Sung., H.M., Uddin, M.J., Mondal S.K.,(2024). Irreversibility of Extreme Precipitation in Global Monsoon Areas Under Multiple Carbon Neutrality Scenarios. The Korean Meteorological Society Fall Conference (2024).

## HONOURS AND AWARDS

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- **Brain Korea 21 (BK21) Scholarship by Republic of Korea (September 2023)**
- **Summer 2024 Research Fellowship – Jeonbuk National University (February 2024).**

## LANGUAGE PROFICIENCY

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- **English**
  - IELTS overall band score 6.0 (July 15, 2023);  
Advanced Public Speaking Course; Phonetics course completed
- **Mother Tongue**
  - Bangla

## REFERENCES

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### **Prof. Dr. Jong-Yeon Park**

Associate Professor,

Department of Earth and Environmental Sciences, Jeonbuk National University, Republic of Korea.

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### **Prof. Dr. Md. Jalal Uddin**

Assistant Professor,

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I certify that the information provided is accurate and complete to the best of my knowledge.